

Jacob Hudnut

Computer Science graduate with a concentration in Robotics and Computer Vision, seeking software development or robotics engineering opportunities where I can apply experience in open-source development, robotics, computer vision, high-performance computing, and technical documentation.

RESEARCH AND PROJECTS

LiDAR Point Cloud Surface Reconstruction

- Designed surface reconstruction and Compression Strategy from a collection of points in 2D space obtained by a LiDAR scan using the RPLiDAR A1 system on the Yahboom Rosmaster X3.

Rensselaer Center for Open Source

- Wrote significant documentation and a web-based ROS2 Iron and Humble install script in an executable environment.
- Acted as Project Lead for a new open source learning platform
- Acted as Project Lead for a project focused on robotics maintenance, documentation, and sample code for RPI robots used in robotics courses.

Highly Parallelized Sobel Edge Detection

- Implemented Sobel Edge Detection Algorithm to run on AiMOS supercomputer, leveraging MPI and Cuda for massive parallelization.

AI-Monitored Hydroponic Garden System

- Designed and built hydroponic garden from scratch components
- Implemented AI computer vision model to detect disease

Distributed Quantum Computing Research

- Researched Quantum Fourier Transform Qubit mapping for Distributed Quantum Computing.

EXPERIENCE

Intern, Data Management and Community Outreach

Lamoille County Conservation District - Morrisville, VT, May 2023 to September 2023

- Organized and compiled spreadsheets and mailing lists, organized disaster relief efforts.
- Engaged with community members through school groups, educating about watersheds and benthic macroinvertebrates in Vermont.

Machine Operator

Hypertherm - Lebanon, NH, June 2022 to January 2023

- Regulated multiple CNC lathe machines simultaneously.
- Demonstrated strong skills in attention to detail, project time management, fast and accurate mental math, spatial reasoning, and visualization
- Gained the necessary knowledge through extensive training required for running EMCO, Citizen, Hardinge, Mori, ABX, and Tugami CNC lathes.

Peer Academic Mentor

Community College of Vermont - Winooski, VT, September 2021 to May 2022

- Provided personalized assistance with a multitude of subjects ranging from introductory writing courses to some of the most advanced coding, science, and math courses offered at the Community College of Vermont.
- Demonstrated and was commended for patience, willingness to listen, attention to detail, and capacity to see from the perspective of others, allowing for consistent identification of misunderstanding and prompt, applicable, easily implementable guidance

Computer Systems Manager

Main Street Kitchens - Hanover, NH, December 2018 to April 2020

- Developed a guidebook to using the store's computer systems
- Provided individualized training for new hires and coworkers in operation of computer systems.
- Kept and audited exhaustive computerized inventory of items, as well as using statistical analysis to identify trends in sales data.

Troy, NY 12180

jacob.hudnut@gmail.com

[LinkedIn](#) | [GitHub](#)

EDUCATION

Bachelor of Science in Computer Science, Robotics and Computer Vision Concentration

Rensselaer Polytechnic Institute-Troy, NY

August 2023 to May 2026

Associate of Science in STEM Studies - Cum Laude

Community College of Vermont - Winooski, VT

January 2019 to June 2023

AWARDS

Student Recognition Award

Academic Honors

HackRPI 2023 Best Sustainability

SKILLS

ROBOT PROGRAMMING

Kinematics, Dynamics, Control, and Sensing

ROS/ROS2

SLAM

COMPUTER VISION AND GRAPHICS

CGAL

Surface Reconstruction

HIGH PERFORMANCE COMPUTING

MPI

Cuda

AI PROGRAMMING

PyTorch/BoTorch

MANUFACTURING

Lathes

Mills

TEAM MANAGEMENT

Training

Project management